

TEAI – A Multi-Module AI System for End-to-End Tea Quality Assessment



Project ID: 25-26J-292





Problem Statement

Context

**TEA MANUFACTURING STILL MAINLY
DEPENDS ON MANUAL INSPECTION AND
HUMAN EXPERIENCE.**

Stakeholders

**TEA TASTERS, SUPERVISORS, QUALITY
OFFICERS, MANAGERS, AND BROKERS
ARE THE MAIN STAKEHOLDERS.**

Observed Issues

**MANUAL CHECKING IS SLOW,
INCONSISTENT, AND MAY MISS
IMPORTANT QUALITY AND
CONTAMINATION PROBLEMS.**

Why It Matters

**THIS CAN REDUCE TEA QUALITY,
INCREASE ERRORS AND COSTS, AND
LOWER SAFETY AND PROFITABILITY.**

Research Gap



Gap

Existing solutions do not provide one integrated AI system to support all major tea manufacturing quality and process decisions.

Main objective

The main objective is to develop an AI-based system that improves tea quality evaluation, process monitoring, safety detection, and decision-making with high accuracy and efficiency.

Gravity of the Problem



Solution summary

We developed an integrated AI-based tea manufacturing support system for quality evaluation, process monitoring, contamination detection, and price prediction.



Key Feature

- tea appearance analysis
- plucking quality grading
- withering prediction
- foreign particle detection
- auction price prediction



Non-Functional

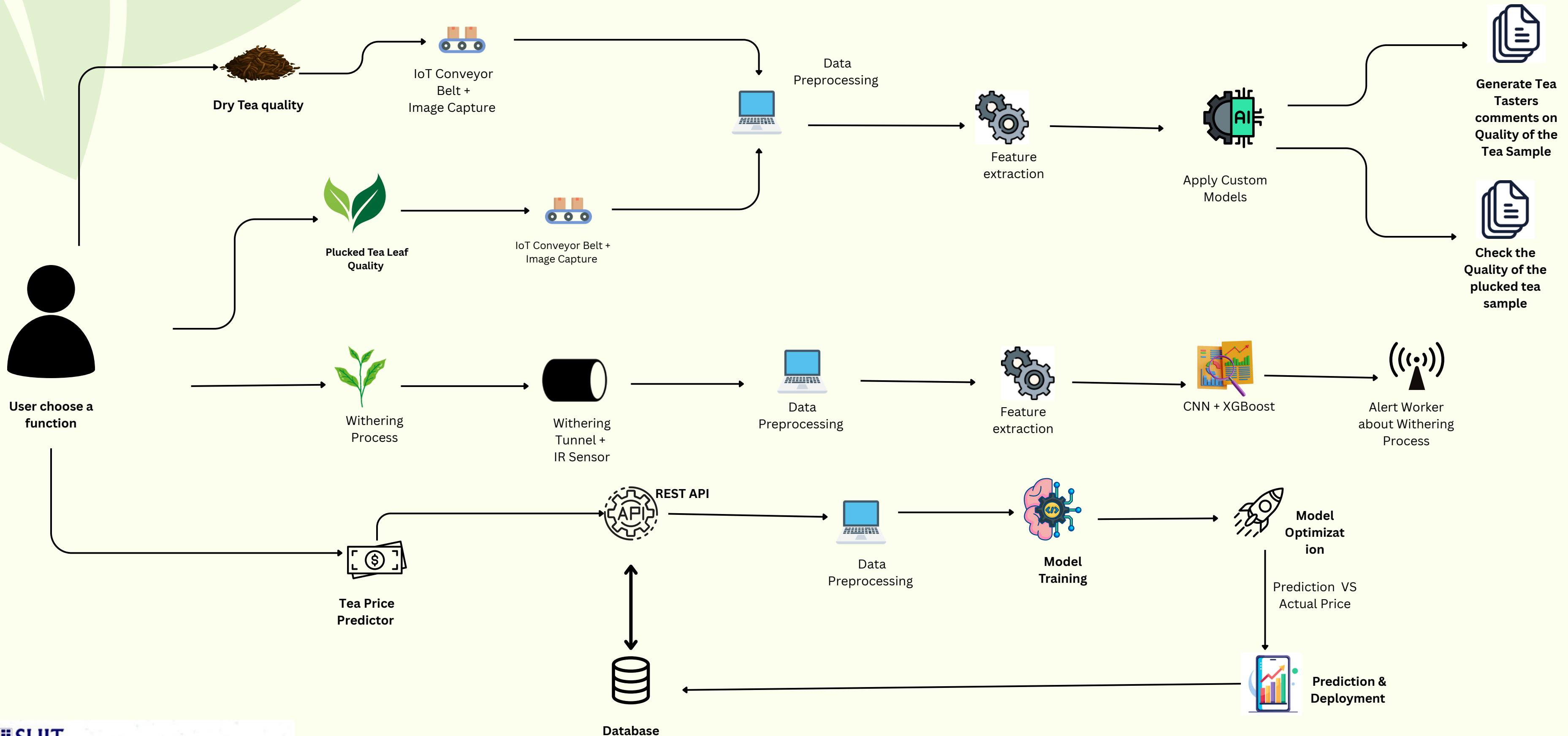
System is designed to be accurate, reliable, and easy to use.



Completion status

The project is over 90% complete.

System Diagram



CONSTRAINTS



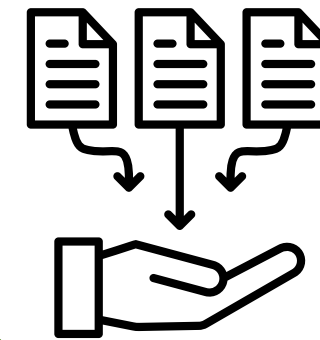
Target users

Tea factories and quality teams benefit from this system.



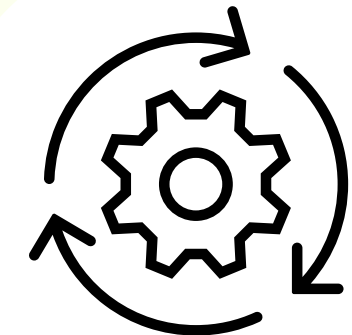
Value proposition

It is faster, smarter, and more consistent than manual methods.



Cost and feasibility

The system is low-cost and practical to implement.



Next steps

It can start with a pilot and later expand to full factory use.



Thank You

